Market Conditions and Price Forecasting for Carriage Services (CSV)

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Can daily trading days be grouped into meaningful market conditions, and can ARIMA forecast the closing price better than a naïve forecast?

Machine Learning for Business · HDip in Data Analytics for Business · CCT College Dublin

Data: Yahoo Finance, ticker CSV · 3 Jan 2017 – 30 Sep 2026 · downloaded 21 Oct 2026

1 Introduction

Carriage Services (CSV) is a US funeral and cemetery company listed on the NYSE. Its daily trading data is both a set of **multivariate records** (open, high, low, close, volume) and a **time series**. We use clustering to find market conditions without labels, and ARIMA to test whether past prices help forecast future prices.

Objectives

- Clean and document 10 years of daily data
- Compare K-Means and Agglomerative clustering (silhouette, Davies–Bouldin)
- Select ARIMA/SARIMA orders with evidence
- Test forecasts out of sample against naïve baselines

2 Data & preparation

2,449	0	6 + 4
trading days	rows removed (no missing / invalid)	original + derived variables

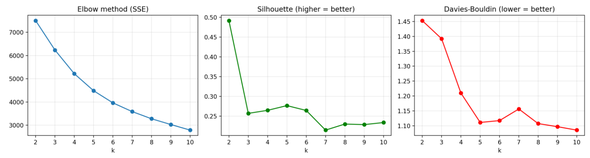
- **Source:** Yahoo Finance via yfinance; dated CSV snapshot; 3 rows checked against the web page.
- **Original:** Date, Open, High, Low, Close, Adj Close, Volume.
- **Derived:** log return, intraday range, 20-day volatility, relative volume.
- **Returns have heavy tails** (excess kurtosis 27.4).



Close price with 50/200-day moving averages, and daily volume

3 Clustering trading days

- **Features:** the 4 derived features, scaled with StandardScaler (Euclidean distance).
- **K-Means** and **Agglomerative (Ward)** tested for k = 2–10.
- **Single / complete / average linkage** scored up to 0.87 silhouette only by isolating 1–10 outlier days, so they were rejected.



K-Means: SSE, silhouette and Davies–Bouldin for k = 2–10

k = 3 chosen: k ≥ 4 gives clusters under 1% of days; DBI favours 3 over 2; at k = 3 both algorithms separate activity and volatility.

k = 3	K-Means	Ward
Silhouette ↑	0.257	**0.312**
Davies–Bouldin ↓	**1.392**	1.481
Stability (ARI, 80% samples)	**0.877**	0.476
Smallest cluster	7.1%	11.0%

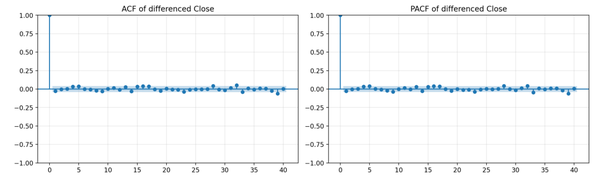


Close price coloured by cluster (0 calm · 1 busy · 2 high volatility)

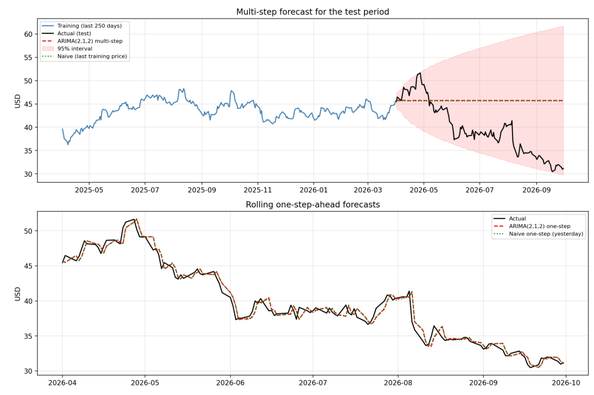
K-Means cluster	Days	Range	20d vol.	Rel. vol.
0 Calm	57%	2.2%	1.8%	−0.40
1 Busy	35%	3.2%	1.8%	+0.35
2 High volatility	7%	6.4%	5.4%	+0.08

4 Time series & ARIMA

- **Target:** daily Close. Train Jan 2017 – Mar 2026, test Apr – Sep 2026 (126 days).
- **ADF:** level p = 0.59 (not stationary); 1st difference p < 0.001, so d = 1.
- **ACF/PACF:** no significant lags 1–27; no spikes at seasonal lags 5–252, so no SARIMA.
- **AIC grid (p, q ≤ 3):** ARIMA(2,1,2) best AIC; ARIMA(0,1,0) best BIC.



ACF and PACF of the differenced Close (training data)



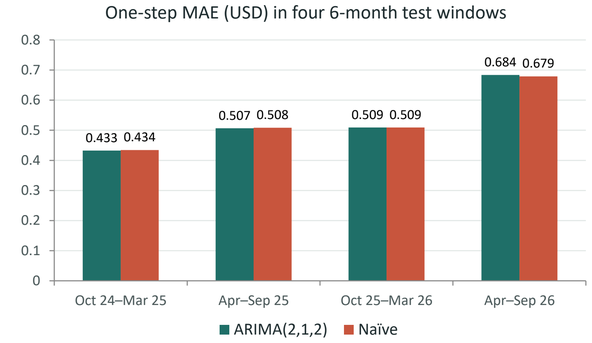
Test period: multi-step (top) and rolling one-step forecasts (bottom)

Test MAE (USD)	ARIMA(2,1,2)	Naïve
One-step ahead	0.684	**0.679**
Six months ahead	7.13	**7.08**

Residual diagnostics (training)

- No autocorrelation left (Ljung–Box p ≥ 0.61)
- Not normal: heavy tails (kurtosis 26.6)
- Variance not constant (H = 2.47, p < 0.01)
- 95% interval covered every test day only because it is very wide

5 Is ARIMA better than naïve?



ARIMA wins 2 windows, loses 2, by at most 0.005 USD: effectively no better than naïve.

6 Discussion & conclusions

- **Three market conditions:** calm (57%), busy high-volume (35%) and high-volatility days (7%). The volatile days cluster in 2018, 2020, 2022 and 2023, although no dates were used.
- **K-Means preferred:** better DBI, far more stable, easier to explain. Ward has the higher silhouette. Separation is weak overall.
- **Forecasting:** the price behaves like a random walk; ARIMA did not beat "tomorrow = today".
- **Link:** residuals are heavy-tailed with changing variance, the same volatility regimes the clusters show, so ARIMA intervals are unreliable.
- **Limitations:** one stock, no external data, k is a judgement, one main split. Not investment advice.

References

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Pedregosa, F. et al. (2011) 'Scikit-learn', JMLR, 12, pp. 2825–2830.

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